ATHARVA ARUN KHAMKAR

DevOps Engineer | Cloud Engineer | AWS | Azure | Kubernetes | CI/CD

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PROFESSIONAL SUMMARY

DevOps and Cloud Engineer with an MSc in Advanced Computer Science (Cloud Computing) from the University of Leeds and two DevOps-focused internships spanning AWS, Linux administration (RHEL, Ubuntu, CentOS) and CI/CD automation. Builds Jenkins and GitHub Actions pipelines that take code from commit to Kubernetes with automated testing, SonarQube quality gates and Docker image builds. Automates provisioning with Ansible and Bash, and used Microsoft Azure to benchmark distributed workloads and produce data-driven compute right-sizing guidance. Proven results include a 60% smaller Docker image, 30% faster container start-up and 80% faster environment setup. Seeking a Graduate or Junior DevOps, Cloud, Platform or Site Reliability Engineer role.

TECHNICAL SKILLS

Cloud Platforms: AWS (EC2, S3, RDS, IAM, VPC, Security Groups, Elastic Load Balancing, Auto Scaling), Microsoft Azure (Virtual Machines)

CI/CD & DevSecOps: Jenkins (Pipeline as Code), GitHub Actions, SonarQube (SAST), GitHub Webhooks, Quality Gates

Containers & Orchestration: Docker, Docker Compose, Multi-stage Builds, Kubernetes (Deployments, Services, Persistent Volumes), Kind

Infrastructure as Code & Config: Terraform, Ansible

Monitoring & Observability: Prometheus, Grafana

Linux & Networking: RHEL, CentOS, Ubuntu, System Administration, SSH, firewalld, cron, Apache, MySQL, Network Troubleshooting

Scripting & Version Control: Bash/Shell Scripting, Python, YAML, Git, GitHub, GitLab

PROFESSIONAL EXPERIENCE

DevOps Intern **Dec 2024 – May 2025**

Radical Technologies, India

- Administered Red Hat Enterprise Linux (RHEL) servers throughout a 6-month hands-on programme, managing users, groups, file permissions, processes and packages (yum/dnf) to keep multi-user systems secure and consistently configured.
- Secured remote server access by configuring SSH and firewalld rules to allow only required services, and resolved connectivity faults using standard Linux networking tools.
- Replaced repetitive manual administration with Bash scripts scheduled via cron, ensuring routine tasks ran consistently without operator intervention.
- Deployed a complete LAMP stack (Linux, Apache, MySQL, PHP) on RHEL from a bare OS: configured Apache virtual hosts, hardened MySQL and exposed only the necessary firewall ports, then validated the live application end-to-end.

Software Engineer Intern (DevOps & Cloud) **Dec 2023 – Jan 2024**

ScaleFull Technologies LLP, India

- Removed manual deployment effort by designing Jenkins CI/CD pipelines with Docker that automated the full build, test and deploy workflow, giving the team faster and repeatable releases.
- Integrated SonarQube static code analysis into the pipeline as a quality gate, catching bugs and code smells before code reached deployment.
- Provisioned and managed AWS EC2 instances for application hosting, configuring Security Groups and IAM roles to control network access and permissions.
- Cut LAMP and WordPress environment setup from a manual, multi-step process to a single automated run by writing Ansible playbooks and Bash scripts.
- Administered Ubuntu and CentOS servers (user management, shell scripting, troubleshooting) to keep development environments stable.

PROJECTS

Distributed Parallel Genetic Algorithm on Microsoft Azure **Jul 2026 – Sep 2026**

MSc Dissertation, University of Leeds | Azure VMs, Distributed Computing, Benchmark Automation

- Identified a gap in cloud benchmarking for distributed parallel workloads and owned the design, build and deployment of a parallel processing system on Azure from scratch, with no existing codebase or template.
- Provisioned Azure VMs and architected the distributed worker system end-to-end, including task decomposition, an inter-worker coordination layer and distributed job scheduling.
- Automated a reproducible benchmarking harness with scripted test runs to measure runtime and throughput across 1–12 concurrent workers, minimising run-to-run measurement variance.
- Achieved consistent horizontal scaling (1.79x speed-up at 2 workers, 4.28x at 12 workers) and identified the cost-efficiency crossover at ~10 workers, beyond which added compute no longer justified the spend.
- Translated results into Azure infrastructure sizing guidance, giving teams a data-driven framework to right-size parallel compute before deployment.

End-to-End CI/CD Pipeline on AWS

Jenkins, GitHub Actions, SonarQube, Docker, Kubernetes, AWS

- Replaced a fully manual release process, where hand-run steps slowed releases and caused inconsistent environments, with an automated, webhook-triggered delivery pipeline.
- Built a Jenkins pipeline triggered on every GitHub commit that runs build, unit tests, SonarQube SAST, Docker image build and registry push, then a rolling Kubernetes deployment, with no manual steps from commit to deployment.
- Added GitHub Actions pull-request checks (linting, tests, static analysis) as a pre-merge quality gate so that only validated code reached the main branch.

Spring Boot Banking Application: Containerisation & Kubernetes

Java Spring Boot, Docker, Kubernetes, Jenkins

- Took on a monolithic Spring Boot application whose oversized Docker image and slow start-up blocked efficient Kubernetes deployment and increased resource cost.
- Re-engineered the image with multi-stage Docker builds that strip build-time dependencies, wrote Kubernetes Deployment and Service manifests, and automated the build and registry push in Jenkins.
- Reduced Docker image size by 60% and container start-up time by 30%, and removed manual release steps, enabling faster pod scheduling and lower resource overhead.

2-Tier Cloud-Native Application on AWS

AWS VPC, Load Balancer, Auto Scaling, Docker Compose, Kubernetes, Flask, MySQL

- Designed and deployed a production-style 2-tier architecture (Flask application tier, MySQL database tier) built to handle variable traffic.
- Provisioned the AWS network stack (VPC, Security Groups, Load Balancer, Auto Scaling Groups) to distribute traffic and scale the application tier automatically.
- Containerised both tiers with Docker Compose, then migrated to Kubernetes with Persistent Volumes for MySQL to retain data across pod restarts.
- Reduced environment setup time by 80% through full containerisation, delivering consistent, repeatable deployments across dev and test and minimising environment drift.

EDUCATION

MSc Advanced Computer Science (Cloud Computing) **Sep 2025 – Sep 2026**

University of Leeds, Leeds, UK

B.Tech Computer Science and Engineering **Dec 2021 – May 2025**

D.Y. Patil Technical Campus, Kolhapur, India

- Graduated in the top 15% of the cohort.

LEADERSHIP & ACTIVITIES

- **Lead,** Linux Club (Year 3), B.Tech.
- **Team Lead,** College Technical Fest, B.Tech.